AUTHENTICATED ENCRYPTION

Until Today: message secrecy and authenticity as separate goals.

Now: We want both at the same time. This builds a SECURE CHANNEL.

Used widely: e.g. TLS 1.3 & VPNs.

First thing we need is a definition!

This corresponds to a SKE that hides

the msg but also ensures nobody can tamper with msg.

All what we studied fails! E.g.

$$c = (c_1, c_2) = (r, F_k(r) \oplus m)$$

$$\tilde{c}_2 = c_2 \oplus 0^{n-1}1$$

$$= F_k(r) \oplus (m \oplus 0^{n-1}1)$$

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An example of a MALLEABILITY ATTACK:

by changing CTX C we obtain an encryption $\tilde{C}$ of msg $\tilde{m}$ related to m in a non-trivial way.

Non-malleable: Above attack are impossible in general sense.

Many real-world applications:

- Digital auctions.

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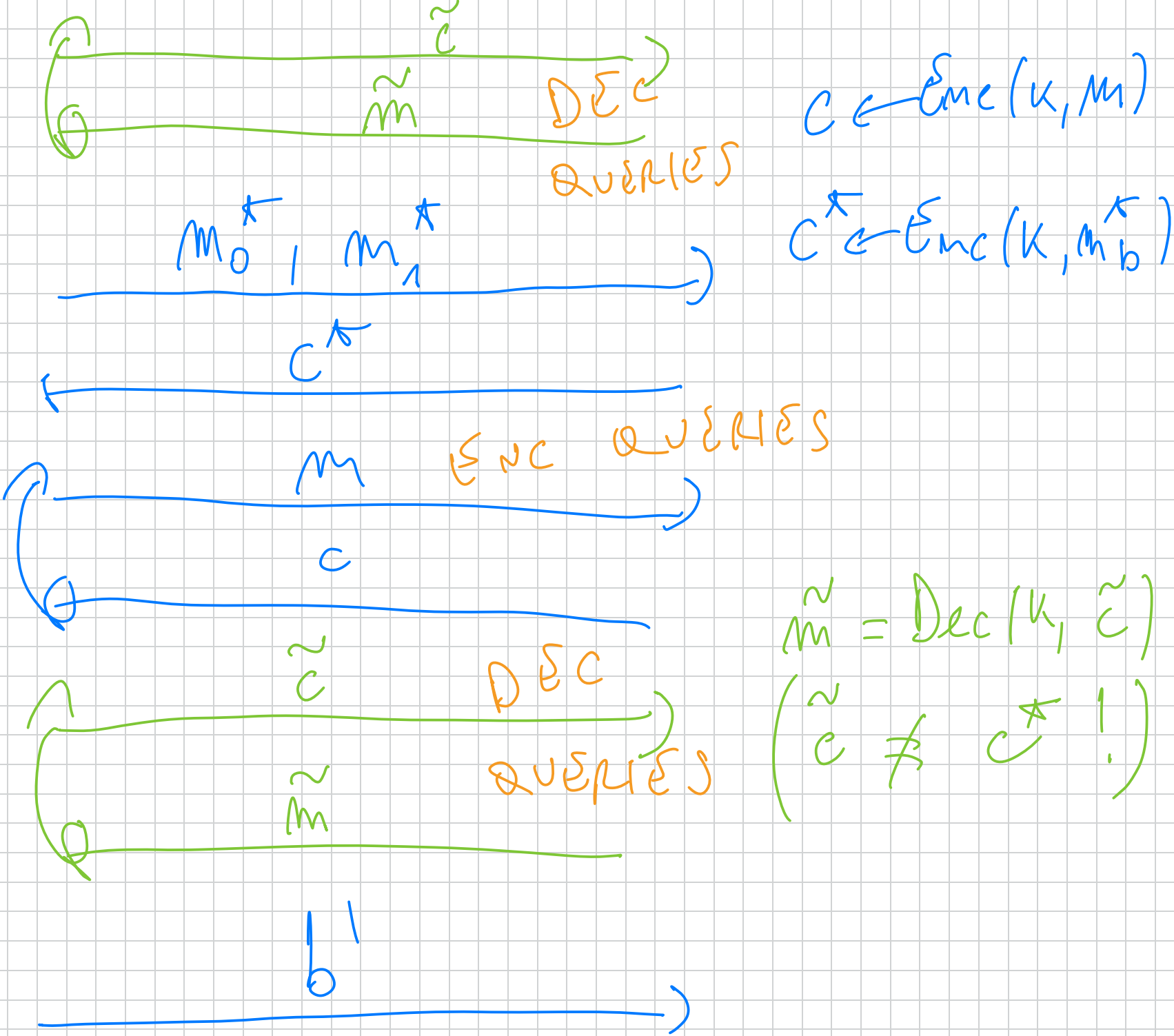
- Bleichenbacher attack against TLS - Malleability of RSA (see a later lecture).

DEF (CCA security) Same as CPA security but with a different game.

$$\text{GAME}^{\text{CCA}}_{\Pi,\mathcal{A}}(\lambda, b)$$



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Exercise The above considered SKE

$(c_1 = r ;\ c_2 = F_k(r) \oplus m)$ is

not CCA-secure.

$$c^* = (c_1^*, c_2^*)$$

$$\tilde{c} = (c_1^*, c_2^* \oplus 0^{n-1} 1)$$

$$\tilde{m} = m_b^* \oplus 0^{n-1} 1$$

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We'll see a construction based on 2 properties:

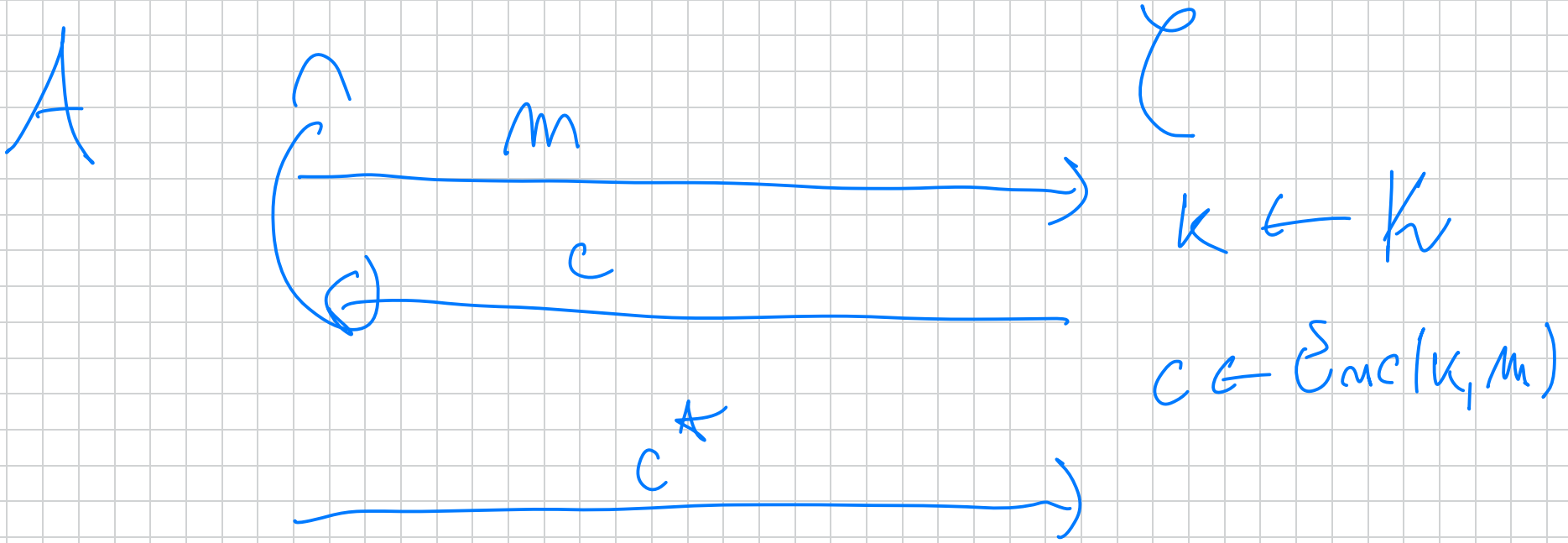
1) CPA security.

2) AUTHENTICITY (a new property).

Let $Dec : K \times C \to M \cup \{\perp\}$

where $\perp$ means "DECRYPTION ERROR".

$Game^{auth}_{\Pi,A}(\lambda)$



A wins:

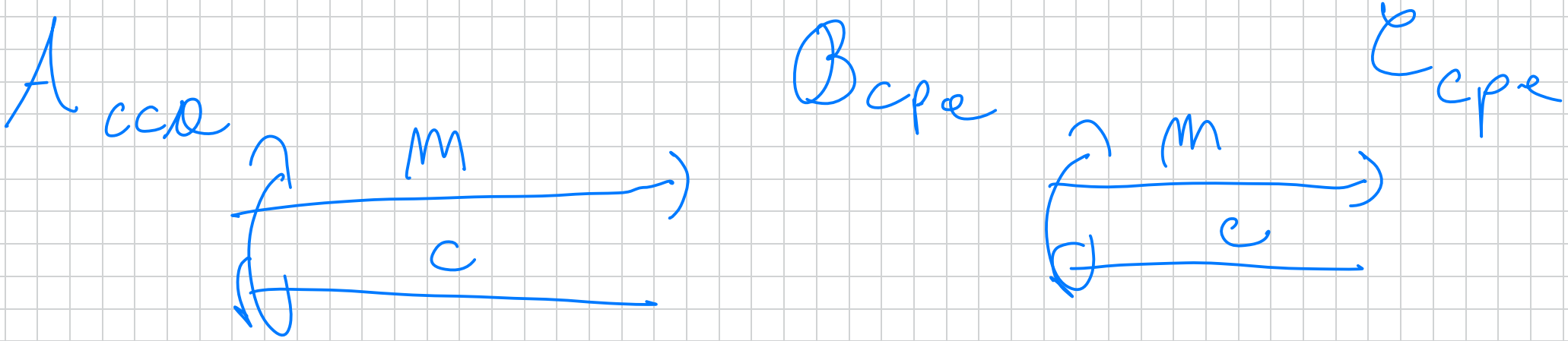
- $c^* \notin \{c\}$
- $Dec(k, c^*) \neq \perp$

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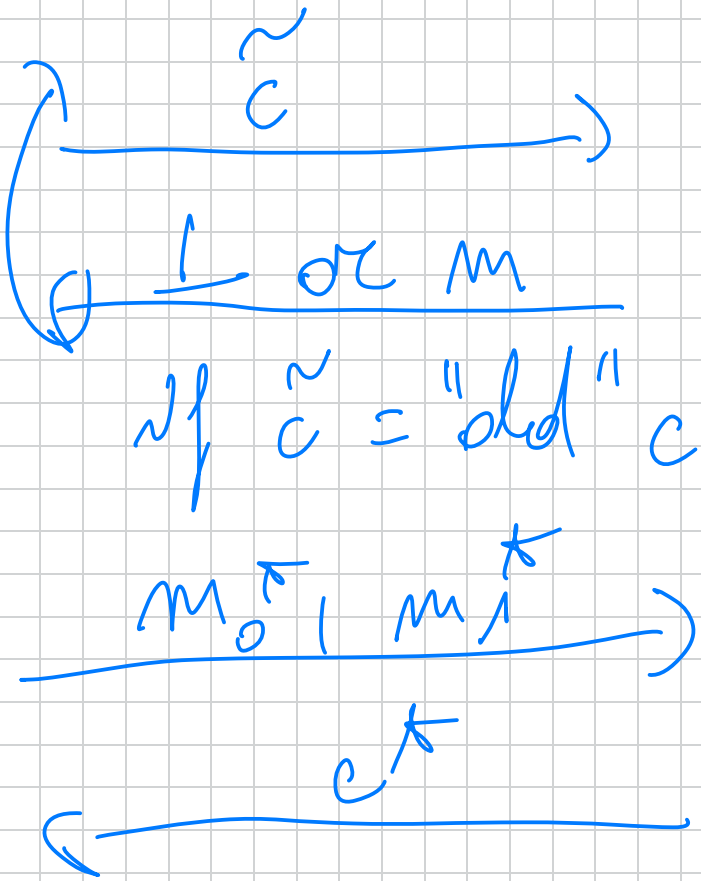
THM CPA + AUTH $\Rightarrow$ CCA.

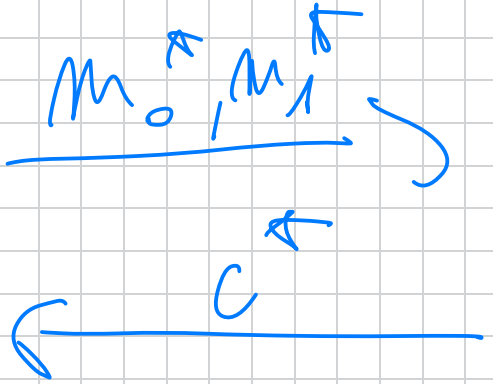
Intuition: Since AUTH holds, decryption queries are useless! So we can reduce CPA to CCA.

Sketch of the reduction:



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Let BAD be the event: $\exists \tilde{c}_i$ s.t. $\tilde{c}_i$ passes and VALID.

$$\Pr[A \text{ wins}] \leq \underbrace{\Pr[A \text{ wins} \wedge BAD] + \Pr[A \text{ wins} \wedge \overline{BAD}]}_{\varepsilon_{\text{enc}}}$$

$$\leq \varepsilon_{cpa}$$

□

How to get 1) + 2) ? Combine SKE with MAC. There are many ways to do it:

A) $c \leftarrow Enc(k_1, m)$; $\tau = Tag(k_2, m)$

$$c' = (c, \tau)$$

ENCRYPT-AND-AUTHENTICATE.

CCA secure or not ?! This is NOT ALWAYS secure : Consider Tag that

outputs $m[0] \| \tau'$ where τ' is a tag on m.

B) $c \leftarrow Enc(K_1, m \| \tau)$

$\tau = Tag(K_2, m)$

Secure or not? No. Despite this is used in TLS 1.2.

Exercise: Break it!

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c) Here is the correct way:

$$c \leftarrow \mathcal{E}_{MC}(K_1, m)$$

$$\tau = T_{eff}(K_2, c)$$

$$c' = (c, \tau) \; .$$

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